

Nirma University
Institute of Technology
Semester End Examination (IR), January - 2022
Master of Computer Application, Semester - I
3MCA103 Operating System

**Roll/
Exam No.**

**Supervisor's Initial
with date**

Time: 2 Hours

Max Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

Answer the following question

Q-1 What is an operating system? Give any two examples (names) of operating system. List the resources of a computer system those are managed by an operating system. Mention the main objectives of operating system and explain them in brief. **[7]**
CO1, BL1

Q-2 What is a process? With respect to process management by an operating system, explain the following terms: **[7]**
CO2,3 BL2

- mutual exclusion
- critical section
- Turnaround time

Q-3 What is a thread in context of operating system? Explain ULT (User Level Thread) and Kernel Level thread discussing advantages and disadvantages of both. **[7]**
CO2,3, BL3

Q-4 Define deadlock. Describe necessary and sufficient conditions leading a system to deadlock. Demonstrate deadlock modelling of two processes and two resources with a diagram. **[7]**
CO2,3 BL4

Q.5 Consider, a process has 8 virtual pages and is assigned a fixed allocation of four page frames in main memory. The pages are referenced in following order : **[7]**
CO2,4,BL6

0 2 3 1 4 2 5 6 0 1 3 2 4 7 1 2

If the main memory frames are initially empty, determine the number of page faults that will occur with the following page replacement algorithms.

(i) FIFO (ii) LRU

Illustrate the procedure using suitable diagram.

OR

Q.5 What is fixed partitioning and dynamic partitioning scheme in context of memory management? Also discuss the advantages and disadvantages of both in detail. **[7]**
CO2,4 BL5

Q.6 Differentiate a **multiprocessor system** with a **multicore system**? [7]
 CO1,3, Discuss SMP (Symmetric Multi-processor) architecture in detail using a
 BL2 suitable neat diagram. Also describe the potential benefits of SMP over
 uniprocessor architecture.

Q-7 List techniques for performing I/O (Input/Output). Discuss DMA in [8]
 CO2,3 BL5 detail using suitable block diagram. Also explain the various ways in
 which DMA mechanism can be configured.

OR

Q-7 Discuss **SPN (Shortest Process Next)** and **SRT (Shortest Remaining [8]
 CO2,3 BL6 Time)** processor scheduling algorithm. Also illustrate the execution
 pattern (timeline chart) diagram for SPN and SRT processor scheduling
 algorithms for the following set of processes given their arrival times
 and service times.

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2